

Arnav Raj Metrani

Mohali, India | arnavmetrani@gmail.com | github.com/ArnavMetrani | arnavmetrani.github.io

EDUCATION

Indian Institute of Science Education and Research, Mohali
Bachelor of Science and Master of Science in Physics (BSMS Physics)

Mohali, India
Jan 2022 — June 2026

PROJECTS

Studying Negative Group Velocity and Pulse Distortion by a Cold Atom Ensemble

June 2025 —

Dr. Mathilde Hugbart- Rb1 team, Cold Atoms group at Institut de Physique de Nice (INPHYNI)

Nice, France

- Worked towards theoretically and experimentally demonstrating negative group velocity in a Rubidium-85 cold atom setup, as well as maximizing the fractional advance for minimal distortion.
 - Performed a literature review of fractional advances obtained in different media.
 - Programming work: Modelled the pulse reshaping after distortion by the atomic cloud, and calculated the ideal experimental parameters for the setup to maximize fractional advance with minimal distortion.
 - Theoretical work: Derived expressions for group velocity through different approaches and with respect to setup parameters, used the transfer function approach for obtaining the output electric field for different frequency spectrums.
 - Experimental work: Characterized optical components (AOM and EOM) to determine the best approach for generating optical pulses within the experiment constraints.
 - Implemented a saturation absorption spectroscopy setup (as training).

Physical Implementations of Quantum Key Distribution Protocols

May 2024 — July 2024

Professor Varun Raghunathan- Indian Institute of Science, Bangalore

Bangalore, India

- Assisted in the analysis and physical implementation of the T-12 protocol and the MDI-QKD protocol.
 - Theoretical work: Analysed BB-84, T-12, MDI-QKD protocols, performed SKR and QBER analysis, analysis of physical components used, characterization of beamsplitter operations, and characterization of single photon and coherent pulses.
 - Programming work included streamlining the instruments' calibration processes via Python and LabVIEW, wrote code for SKR and QBER analysis for T-12 protocol, and analysis of MDI-QKD scenarios via Python.
 - Experimental work included running the setup (for T-12 protocol and MDI-QKD protocol) for calibration purposes and for obtaining real-time experimental data.
- [Project report and code](#)

Introduction to Quantum Algorithms and Quantum Error Correction

May 2023 — July 2023

Professor Kuntal Roy- Indian Institute of Science Education and Research, Bhopal

Online

- Studied various Quantum Algorithms (Quantum Teleportation, Deutsch-Jozsa, Bernstein-Vazirani, Grover search, Simon's problem, Shor's code, Shor's algorithm), and applications of Quantum Fourier Transform such as QFT addition.
- Wrote and executed quantum circuits on IBM-Q's public quantum computers (written on Qiskit).
- [Project report and code](#)

Introduction to Special Relativity

June 2023 — August 2023

Professor Jasjeet Singh Bagla- Indian Institute of Science Education and Research, Mohali

Mohali, India

- Studied Lorentz transforms, Minkowski geometry, paradoxes in SR, four-vector formulation, analysis of accelerating frames including Rindler frames, electrodynamic effects, and optical effects in SR.

COURSE WORK

- **Physics Coursework:** Waves and Optics (A.P. French), Thermodynamics (Callen), Quantum Mechanics (Griffith), Electrodynamics (Griffith+Thompson), Classical Mechanics (Goldstein), Differential Equations, Advanced QM (Perturbation theory - Zettili), Stat. Mech. (Callen), Foundational QM (Nielsen and Chuang), Solid State, Nuclear Physics, Atomic and Molecular Physics (Bransden), QCQI (Nielsen and Chuang), Nonlinear Optics, General Relativity, Physics of Fluids, Introduction to Quantum Optics
- **Math Coursework:** Real Analysis in One variable, Introduction to Differential Geometry, Introduction to Differential Equations, Probability Theory (Ross), Theory of Computation (Sipser), Linear Algebra and Group Theory (Palash Pal)
- **Lab Coursework:**
 - PHY211 and PHY311- Basic optics interference experiments: Thin-film interference, single slit diffraction, Fabry-Perot interferometer, Newton's rings
 - PHY312- Logic gates, flip flops, registers and counters, clock circuits, and programming with Arduino.
 - PHY411- Spectrum analysis of radioactive elements and study of decay processes

SKILLS

- **Programming Languages:** Python, Java, Mathematica, LabVIEW. Design: L^AT_EX, Typst, Markdown, Inkscape
- **Electronics:**
 - Waveform generators (Agilent, Tektronix), oscilloscopes (BK Precision, LeCroy), RedPitaya chip,
 - Laser diode software (PurePhotonics PPCL550),
 - Beam profiling software ([GaussianBeam](#) by Jérôme Lodewyck, [Beam Profiler software](#) by DataRay),
 - Power meter (Thorlabs PM 1000D) and relevant software ([Optical Parameter Monitor](#) by Thorlabs),
 - SPAD operation software and TCSPC software by IDQuantique.
- **Experimental techniques:** Optical alignment, beam coupling, HOM visibility determination, using optical components (polarization analyser, AOM, EOM)

OTHER ACTIVITIES

Talks and Presentations

- ([Extended](#)) [Introduction to Quantum Algorithms](#) (Student-hosted)
- [Quantum Bayes' rule affirming consistency in measurement inferences in quantum mechanics](#) (Seminar course presentation)
- [Quantum circuits cannot control unknown operations](#) (Course presentation)
- [Quantum Cryptography](#) (Student-hosted)
- [Self-Focusing in Nonlinear Media](#) (Course presentation)

Science blog

Oct 2022 —

- Written 10 articles on various topics ([Site link](#)).

Co-convenor of IISER Mohali's Physics Club

Oct 2022 — Oct 2023

- Demonstrations in IISER-M Foundation Day 2023, physics club organiser in 2024 edition.
 - Workshops on Special Relativity, QCQI, CV and Email writing, and PhD applications.
 - Organised 10 talks and a panel talk on Summer Internships.
 - Hosted Grand Canonical Ensemble in 2023. (The Physics Club's annual week.)
-